CricketLens

Stage 3 / Session 3 | Full-history machine learning & held-out 2026 evaluation

Executive outcome. Both models were trained using eligible IPL records beginning in 2008. After validating algorithm choices on 2025, we retrained model weights through 2025, evaluated once on 2026, and then made separate deployment refits through 2026. These are real IPL evaluation results, not sample forecasts.

The sequence - why it matters

Phase	Allowed data	Purpose
Develop	2008-2024	Train full-history candidates
Select	2025	Choose candidate by validation Brier score
Holdout fit	2008-2025	Freeze selected algorithm and fit all pre-2026 records
Independent test	2026	Record untouched-season probabilities and outcomes
Deployment refit	2008-2026	Train final model weights for future matches

2026 results compared with constant-rate baseline

Probability event	Test rows	Model Brier	Baseline Brier	ROC-AUC
Wicket in next over	2,581	0.2005	0.2049	0.589
Batter scores 30+	1,113	0.1853	0.2096	0.709
Batter scores 50+	1,113	0.1179	0.1291	0.729

Brier Score and log loss: lower is better. ROC-AUC: higher is better, 0.5 approximately random ranking. Constant baseline forecasts the 2008-2025 event rate for every 2026 observation.

Interpretation. All three forecasts beat constant-rate baselines on the 2026 holdout. The wicket model improvement is small and does not prove that future individual wickets can be predicted reliably.

Model selection and sample accounting

Model choices frozen using 2025 only

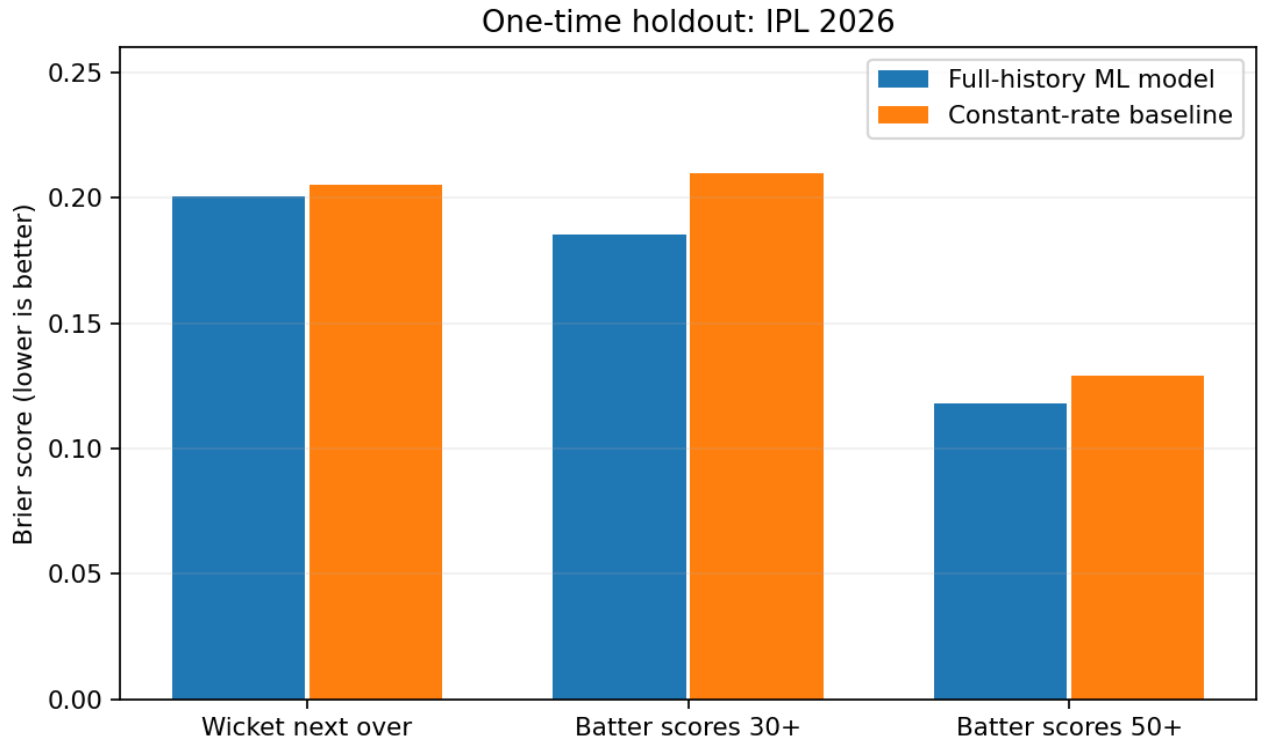
Wicket: logistic context all history. Selected 2025 Brier 0.197057. Full-history Logistic Regression with current score, wickets, completed overs, last-over movement, innings, teams, venue and phase.

Batter: forest full history. Selected 2025 averaged Brier 0.153097. One three-class Random Forest with player form, opposition and venue history, plus match context. Probabilities are combined as P(30+) = P(30-49) + P(50+).

Number of examples at each stage

Dataset slice	Next-over wicket	Batter innings
2008-2024 initial train	39,019	16,231
2025 validation	2,561	1,085
2008-2025 holdout model fit	41,580	17,316
2026 independent test	2,581	1,113
2008-2026 final model fit	44,161	18,429

Brier scores: measured versus baseline

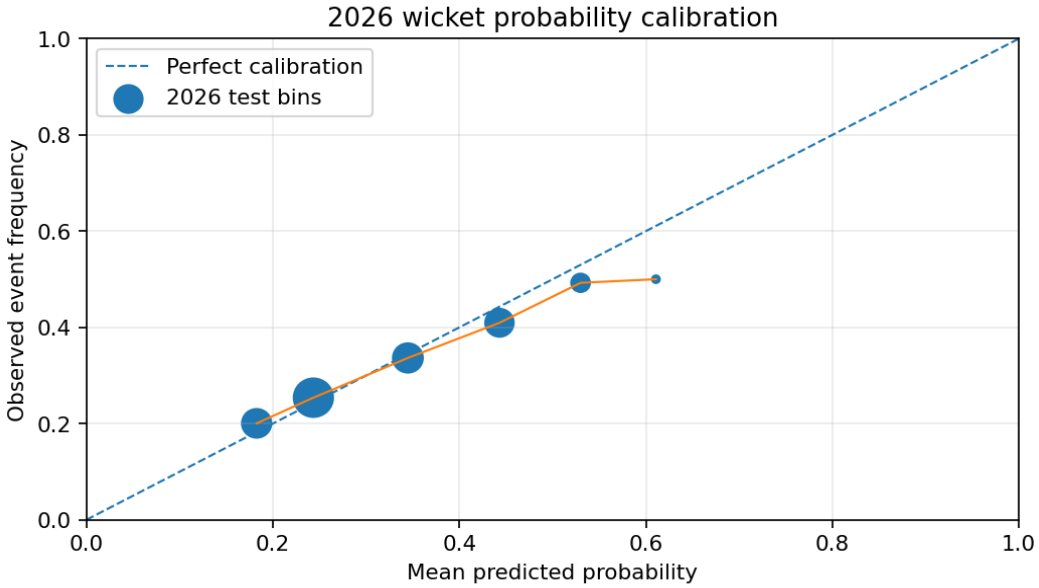


The 2026 holdout results refer to models trained through 2025. Models refitted through 2026 cannot be validly tested on 2026 again.

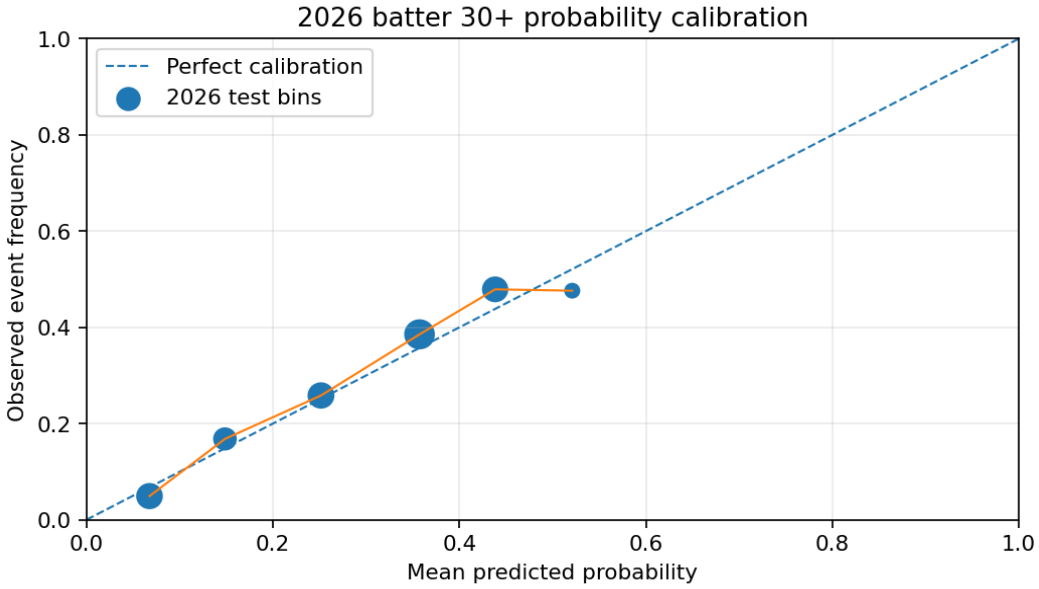
Calibration and reliability

A reliability diagram compares the probability assigned to an event against the fraction of observed events. A perfectly calibrated predictor would follow the diagonal. Sparse bins can be unstable.

Wicket in next over



Batter 30+ runs



Assumptions, limitations and next steps

What these predictions mean

Wicket: after a completed over, estimate the chance of at least one batting-team wicket in the next scheduled over. Runs and wickets in that future over must not become input features.

Batter: before a match, estimate 30+ and 50+ milestone chances **conditional on an observed batting innings**. Playing-XI selection and whether a batter actually bats are not currently modeled.

Exceptional matches and the historical cutoff

• D/L-marked matches are preserved in the historical dataset but excluded from these initial training cohorts. No-result batting innings were also excluded.

• Super Over deliveries are excluded; eligible regulation-innings samples from Super Over matches remain included.

• Short or disrupted upcoming overs without observed wickets are censored and excluded, while witnessed wickets in short next overs count as positive cases. This may introduce selection bias.

• Batter recent-form features for a 2026 prediction may use results from earlier 2026 dates only. The model weights themselves were fixed at the end of 2025.

• Full-history training can dilute recent patterns. We deliberately used 2008 onward for this experiment; this was a project decision, not a claim of guaranteed improvement.

Exactly which files to use

Held-out test: `wicket_holdout_train_through_2025.joblib`, `batter_holdout_train_through_2025.joblib`.

Future predictions: `wicket_deployment_through_2026.joblib`, `batter_deployment_through_2026.joblib`.

Reproduce: run `python -m cricketlens.stage03_full_history` from the project root after installing dependencies. Processed CSVs and models are local-only; do not push generated data to GitHub.

Viva questions we must be able to answer

Why is 2026 a valid test before, but not after, retraining? Why does P(50+) never exceed P(30+)? What does the Brier Score mean? How do we avoid future information in feature engineering? Why might including 2008-2017 hurt? How are D/L and Super Overs handled? What happens if the player does not bat?

Stage status: Stage 3 models are internally evaluated and refitted; dashboard, API, real-time verification and prediction tracking remain Stage 4/5 work.